

He (He I). Strongest classified transitions below 2 μm (NIST): 53 levels, 70 lines.

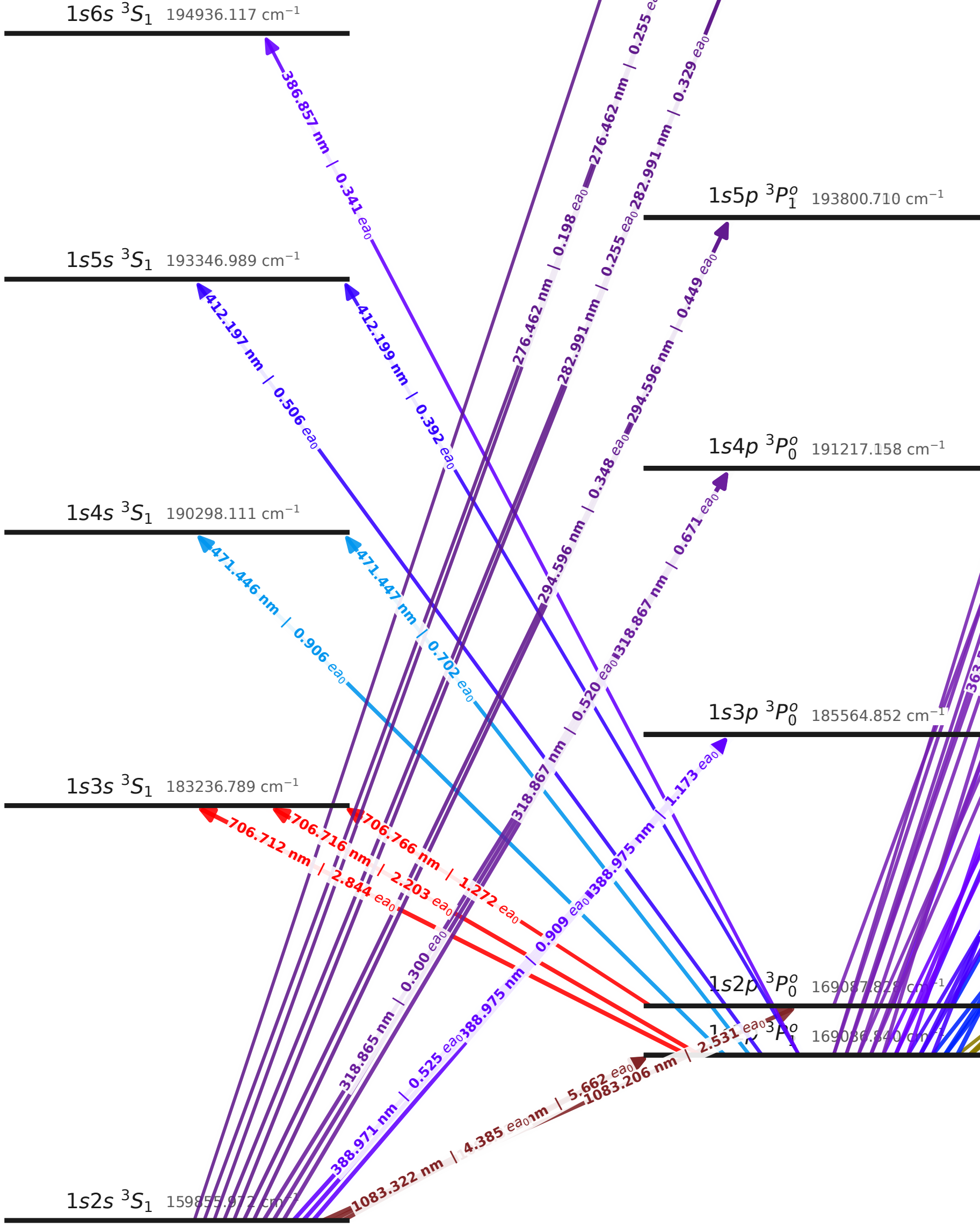
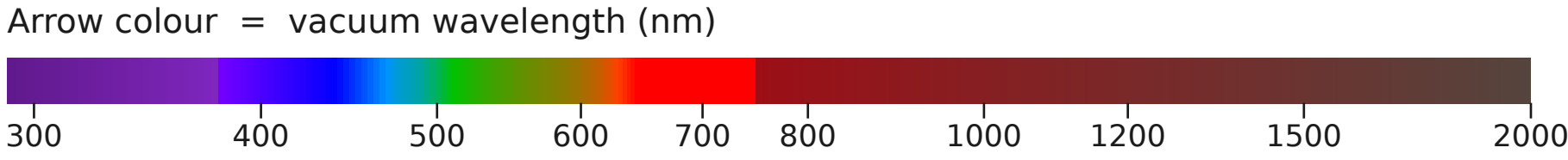
He⁺ ionisation limit $198310.67 \text{ cm}^{-1} = 24.58739 \text{ eV} (50.426 \text{ nm})$

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$

Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e J \rangle$ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$1s^2 \ ^1S_0 - 1s10p \ ^1P_0^o$	50.7058	-	5912.3895	0.055	$1.593e+07$	-	-	-
$1s^2 \ ^1S_0 - 1s9p \ ^1P_0^o$	50.7718	-	5904.7030	0.065	$2.183e+07$	-	-	-
$1s^2 \ ^1S_0 - 1s8p \ ^1P_0^o$	50.8643	-	5893.9616	0.078	$3.103e+07$	-	-	-
$1s^2 \ ^1S_0 - 1s7p \ ^1P_0^o$	50.9998	-	5878.3031	0.095	$4.622e+07$	-	-	-
$1s^2 \ ^1S_0 - 1s6p \ ^1P_0^o$	51.2099	-	5854.1944	0.121	$7.317e+07$	-	-	-
$1s^2 \ ^1S_0 - 1s5p \ ^1P_0^o$	51.5617	-	5814.2487	0.160	$1.258e+08$	-	-	-
$1s^2 \ ^1S_0 - 1s4p \ ^1P_0^o$	52.2213	-	5740.8070	0.227	$2.463e+08$	-	-	-
$1s^2 \ ^1S_0 - 1s3p \ ^1P_0^o$	53.7030	-	5582.4162	0.360	$5.663e+08$	-	-	-
$1s^2 \ ^1S_0 - 1s2p \ ^1P_0^o$	58.4334	-	5130.4951	0.729	$1.799e+09$	-	-	-
$1s2p \ ^3P_0^o - 1s4d \ ^3D_3$	447.2729	447.1474	670.2674	2.756	$2.458e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s4d \ ^3D_2$	447.2744	447.1489	670.2652	2.017	$1.843e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s3d \ ^3D_2$	587.7242	587.5614	510.0903	2.976	$1.767e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s3d \ ^3D_3$	587.7243	587.5615	510.0903	7.042	$7.071e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s3d \ ^3D_1$	587.7254	587.5625	510.0894	2.976	$2.946e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s3d \ ^3D_2$	587.7269	587.5640	510.0881	5.154	$5.302e+07$	-	-	-
$1s2p \ ^3P_0^o - 1s3d \ ^3D_1$	587.7595	587.5966	510.0598	3.437	$3.928e+07$	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.